KRMU PlaceIQ

Smart placement management system

Project report: how it was designed, built and tested, and the technologies used

Prepared for K.R. Mangalam University, Gurugram: Career Development Cell demonstration

Date 31 October 2026

Deliverables A working web application as a GitHub-ready repository, and this report.

What is in this report

- 1. What the system does and why it was built
- 2. The reference application and how this build relates to it
- 3. Data: what is official, what is sample
- 4. Architecture and technology stack
- 5. Data model and algorithms (readiness, matching, adaptive quiz)
- 6. Screens and design decisions
- 7. Testing
- 8. Deployment on Vercel or any static host
- 9. Limitations and the path to production
- Appendix: sources and data quality notes

1. Overview

PlaceIQ is a placement management system for a university Career Development Cell (CDC). It brings together the three jobs a placement team does every season: reporting results, running campus drives, and preparing students. It runs in a web browser, needs no installation and no server, and can be demonstrated from a single link or a single file.

Objectives

- Show official KRMU placement results in one clear dashboard, with the data problems in the published numbers flagged instead of hidden.
- Let a placement officer create drives, see which students are eligible for each one, rank them by fit, and move applicants through the hiring stages.
- Give each student a readiness score, the drives that suit them, the reasons they are blocked from others, and the skills to build next.
- Include an adaptive aptitude round whose difficulty follows the student's answers, and feed its result back into readiness.
- Allow the cell to replace the sample students with its own records by pasting a CSV.

Users and screens

User	Screen	What they do
Placement officer / head	Dashboard	Reads official figures and working-record analytics
Placement officer	Drives	Adds drives, reviews ranked eligible students, adds applicants, updates stages
Placement coordinator	Students	Searches and filters the cohort, inspects readiness and skill gaps, imports CSV
Student	Student portal	Sees readiness, matched drives, applications and next skills
Student	Aptitude round	Takes the adaptive test; the score updates the profile

2. The reference application

The reference supplied was **AdaptIQ**, an adaptive learning and diagnostic tool for school Mathematics and Science, hosted on Vercel at adaptive-quiz-tool.vercel.app. Reading the page returned only its metadata: the title ("AdaptIQ: AI-Powered Adaptive Learning & Diagnostic Tool (SDG 4)"), a description of an adaptive platform for under-resourced schools, and an indigo theme colour (#4F46E5). The application's inner screens and code could not be read, so PlaceIQ does not copy it.

What PlaceIQ takes from the reference is the approach: a self-contained web app on static hosting, a diagnostic that adapts to the user, and a restrained single-accent visual identity. The domain, data model, screens and code are original to this project.

3. Data: official and sample

KRMU publishes aggregate results, not student-level records. The brief was to build on the university's placement records page (krmangalam.edu.in/placements/records). That page and the two CDC annual reports it links were read, and their figures are built into the dashboard. Student, drive and offer records are generated sample data, and the app says so in a banner on every screen.

Item	Source	Status in the app
Placement rate, offers, PPOs, internship offers, recruiters, median package, sector split, package distribution (2025-26)	KRMU placement records page and CDC annual report 2025-26	Official, shown on the dashboard
School-wise eligible, placed, max, average and median CTC (2024-25, 11 schools)	KRMU placement records page	Official, shown as a table with discrepancy flags
Median package by year (2023-24 to 2025-26)	KRMU placement records page	Official, shown as a chart
816 placed of 1,015 eligible, 81% reported (2024-25)	CDC annual report 2024-25	Official, used as a cross-check
Recruiter names (Paisa Bazaar, Ericsson, Red Bull, HDFC Bank, Autodesk, Gaur & Associates and others)	Recruiter lists on the placement page and 2024-25 report	Real names used for sample drives
Roles, CTCs, dates and eligibility rules of drives	Invented for demonstration	Sample
120 students, their CGPA, skills, applications and offers	Generated by a seeded random generator	Sample until a CSV is imported

Why sample students are needed

A placement system is only useful with individual records, which the university does not publish and which are personal data. The generator creates 120 students across six programmes with plausible CGPA, skills, projects, internships and applications, so every feature can be shown working. A fixed random seed makes the sample identical on every load. The Students tab accepts a CSV so the real cohort can replace it.

Data quality problems found in the published figures

These are shown inside the app under "Check before you quote these numbers" and should be confirmed with the CDC before any figure goes into an official document.

- Seven of eleven schools state a placement rate more than 3 points from placed divided by eligible. The largest gaps: Agricultural Sciences 86% stated against 50% computed, Legal Studies 91% against 70%, Medical and Allied Sciences 85% against 70%.
- School of Basic and Applied Sciences lists an average CTC of 10.5 LPA above its own maximum of 10 LPA. Engineering, Education and Basic and Applied Sciences carry the same CTC trio, which looks like a copy error.
- Highest package is Rs 56.6 LPA in the "At a Glance" section but 34 LPA in both annual reports. Average is 4.25 LPA in the 2024-25 report and 6 to 7 LPA in 2025-26.
- The page introduction refers to the 2025-26 cohort, while the school table is labelled 2024-25.
- The lowest band of the 2025-26 package distribution is labelled "Above 1 LPA". The app shows it as 1 to 2 LPA and says this is an interpretation.

4. Architecture and technology stack

PlaceIQ is deliberately a client-side application with no framework, no build tool and no backend. The source is 26 JavaScript modules and about 2,000 lines, split into native ES modules by responsibility: data, core logic with no DOM access, UI helpers and one view per tab. That keeps it easy to test, demonstrate, hand over and host anywhere, at the cost of shared storage (see section 9). Dependencies point one way only: data and core never import from the views.

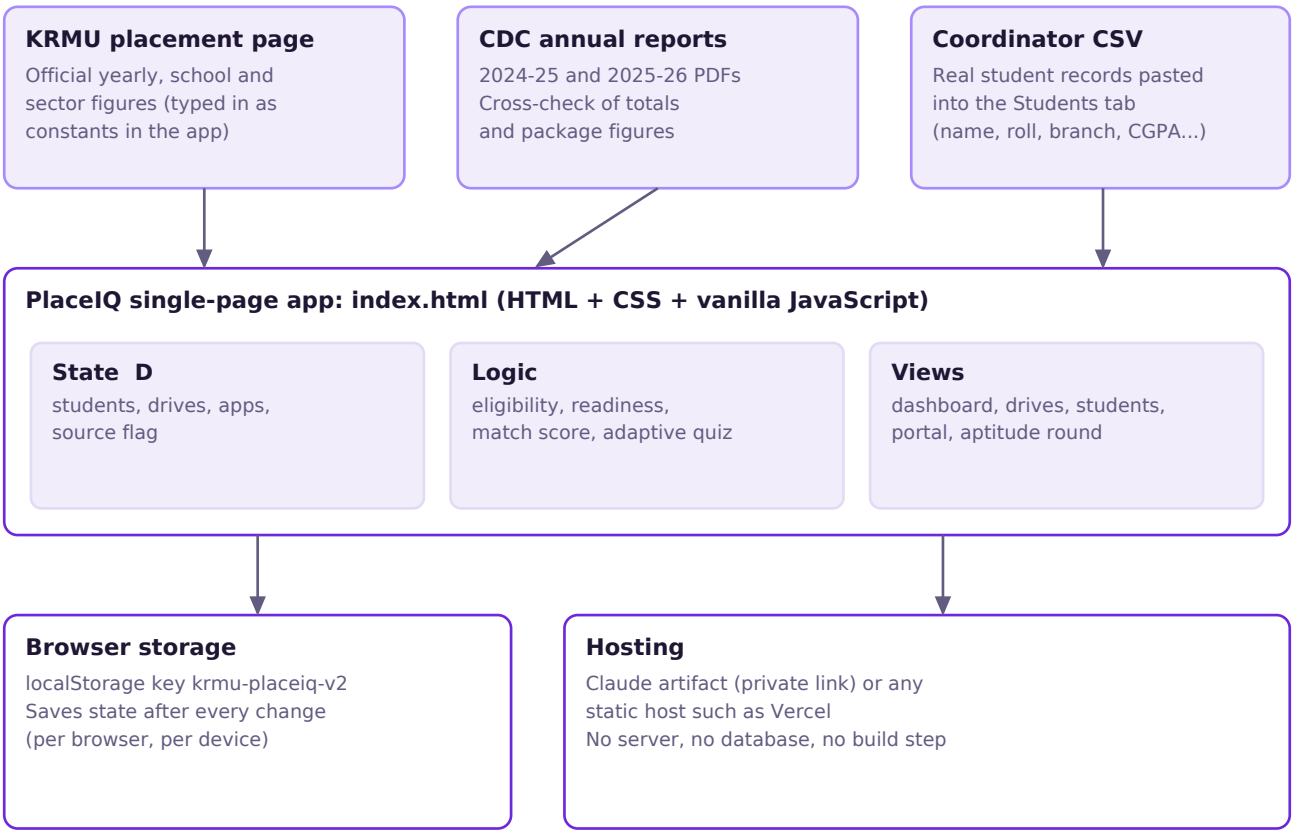


Figure 1. System architecture. Data enters from official sources (typed into the app as constants) or a pasted CSV, flows through one state object, and is saved in the browser.

Technology stack

Layer	Technology	How it is used
Markup	HTML5	Single index.html; semantic regions (header, nav, main, footer), forms, details/summary for the demo walkthrough and the add forms
Styling	CSS3: custom properties, Grid, Flexbox, media queries	Design tokens for colour and type; responsive grids that stack at phone width; prefers-color-scheme for automatic light and dark themes; prefers-reduced-motion respected
Logic	Vanilla JavaScript (native ES modules, no libraries, no bundler)	State object, rendering by template strings, event delegation, sorting, filtering, scoring, CSV parsing, adaptive quiz engine
Routing	URL hash (#dashboard, #drives, #students, #portal, #practice)	Each tab has a bookmarkable address; works on static hosting with no server rules
Persistence	Web Storage API (localStorage)	Key krmu-placeiq-v2; state saved after every change, with try/catch for private-mode browsers
Typography	Google Fonts: Bricolage Grotesque, IBM Plex Sans, IBM Plex Mono	Display headings, body text, and tabular numbers; system font fallbacks if the font host is blocked
Graphics	Inline SVG and CSS bars	Logo mark and all charts are drawn with markup, so no chart library is needed
Data generation	Mulberry32 seeded pseudo-random generator	Reproducible sample cohort of 120 students and their applications
Clipboard	Clipboard API with fallback	Copies the student list as CSV; falls back to a text box if the browser refuses
Hosting	Claude artifact, or any static host (Vercel, Netlify, GitHub Pages)	No server code, so deployment is copying one file
Authoring	Claude Sonnet 5.5 (Anthropic) in an agent workspace	Research, design, code and this report, using web search and fetch, file tools and a shell
Testing	Node.js built-in test runner (49 unit tests); Playwright driving Chrome (25 browser checks)	Unit tests for scoring, quiz engine, CSV and data; scripted walk-through of every tab and action; screenshots at desktop and phone width
CI and hosting	GitHub Actions; Vercel, GitHub Pages or Netlify	Workflow runs both test suites on every push; the app is static files
Report	Python with ReportLab and Pillow	Generates this PDF and embeds the test screenshots

5. Data model and algorithms

State object

Collection	Fields
students	id, name, roll, branch, cgpa, backlogs, skills[], aptitude (0-100), projects, internships
drives	id, company, role, ctc (LPA), minCgpa, maxBack, branches[], skills[], date, status (Open, Upcoming, Completed)
apps	sid (student id), did (drive id), stage (Applied, Shortlisted, Interview, Offered, Rejected)
source	"sample" or "imported"; controls the banner

Eligibility

A student is eligible for a drive when all three tests pass: CGPA at or above the drive's minimum, active backlogs at or below its limit, and the student's programme is in the drive's list. When any test fails, the reason is shown in plain words (for example "CGPA 7.99 is below the 8.0 cut-off") so a student knows what to change.

Readiness score (0 to 100)

Component	Maximum points	Rule
CGPA	35	CGPA divided by 10, times 35
Aptitude	25	Aptitude score divided by 100, times 25
Skills	20	Number of skills divided by 6 (capped at 1), times 20
Projects	10	Projects divided by 3 (capped at 1), times 10
Internships	10	Internships divided by 2 (capped at 1), times 10
Backlog penalty	up to -20	10 points per active backlog, at most 20

The weights are a transparent starting rule, not a trained model. They are editable in one function and should be tuned once the cell has real outcome data.

Match score for a drive

Match = 55% x (share of the drive's required skills the student has) + 45% x readiness. Eligible students are ranked by this number. The skills a student lacks for a drive are listed beside the score, and the most frequently missing skills across open drives for the student's programme form the "Worth building next" list.

Adaptive aptitude round

- Question bank of 18 items: six each at level 1 (easy), 2 (medium) and 3 (hard), covering percentages, ratios, time and work, interest, series, logic, probability, permutations and vocabulary.
- The round starts at level 2 and has 10 questions. A correct answer raises the level by one (to a maximum of 3); a wrong answer lowers it by one (to a minimum of 1). Questions are not repeated; if a level runs out the engine moves to the nearest level that has questions left.
- Score = round(100 x points earned / 28), capped at 100, where a correct answer earns its level (1, 2 or 3) in points. Hard questions answered correctly therefore count for more.
- Every question shows an explanation after the answer. The final score is written to the student's aptitude field, which immediately changes the readiness score.

Student alerts and prep plan

- **Alerts are derived, not stored.** They are computed from the drives, the student's applications and today's date. Only the ids of alerts already read are saved.
- **Kinds of alert:** an offer (good news); a reminder for a drive the student applied to (within 14 days) or was shortlisted or called to interview for (within 30 days); and a new opening the student is eligible for and has not applied to (within 30 days, best five). Urgency is urgent at one day or less and soon at three days or less.
- **Read alerts return when a drive nears.** Alert ids include a time bucket (14, 7, 3, 1 and 0 days to go), so a reminder marked read comes back as unread at the next bucket.

- **Prep plan:** for an eligible student, a checklist built from the drive and the student: apply, research the company, learn each missing required skill (up to four), revise held skills, raise aptitude to 70 if below, update the resume, prepare an introduction, then a technical-round step if shortlisted, mock interview and common questions at interview stage, and a documents check in the last two days. Each step carries a suggested date between today and the day before the drive. Ineligible students see the reasons instead.
- **Demo date.** A date picker overrides today so the feature can be shown on any day; sample drive dates are generated relative to the day the sample is created, so a demo never goes stale.
- **Limit:** alerts show inside the app only. Email, WhatsApp or push messages need a backend (see section 9).

Placement status and dashboard figures

A student is **Placed** when at least one application is at stage Offered (counted once however many offers they hold, matching the CDC's definition), **In process** when an application is active, and **Not placed yet** otherwise. All charts on the working-records half of the dashboard are computed live from the state object, so changing a stage in the portal changes the dashboard.

6. Screens and design decisions

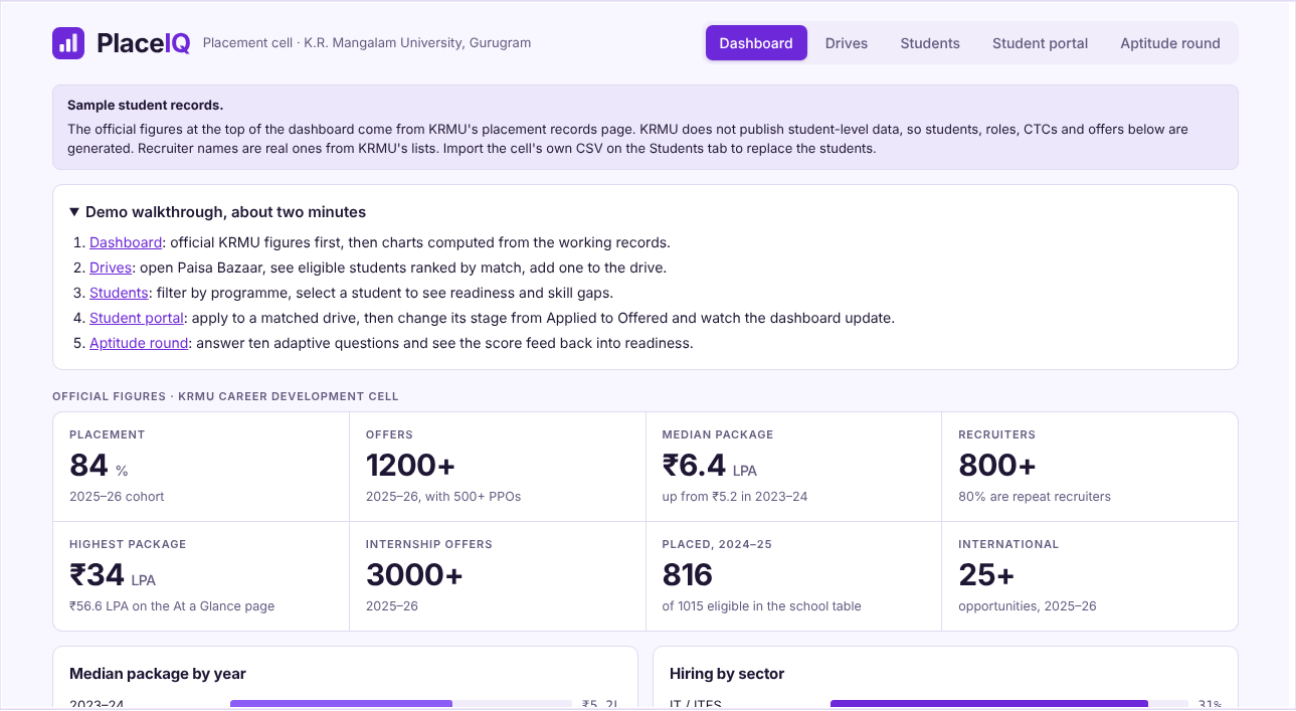


Figure 2. Dashboard: sample-data banner, demo walkthrough, and the eight official headline figures.

School-wise placement, 2024–25 cohort

SCHOOL	ELIGIBLE	PLACED	PLACED ÷ ELIGIBLE	RATE STATED	MAX CTC	AVG CTC	MEDIAN CTC
SOMC School of Management and Commerce	321	266	83%	85%	20	9.8	8.5
SOET School of Engineering and Technology	225	192	85%	85%	34	10.5	9.2
SOLA School of Liberal Arts	135	120	89%	85% differs	14	7.8	7.1
SMAS School of Medical and Allied Sciences	127	89	70%	85% differs	7	5.6	5.2
SOLS School of Legal Studies	93	65	70%	91% differs	12	5.8	5.5
SOED School of Education	29	22	76%	76%	34	10.5	9.2
SBAS School of Basic and Applied Sciences	29	21	72%	82% differs	10	10.5 above max	9.2
SOAD School of Architecture and Design	21	14	67%	86% differs	9	6.8	6.3
SJMC School of Journalism and Mass Communication	18	17	94%	94%	12	7.2	6.7
SOAS School of Agricultural Sciences	14	7	50%	86% differs	6.5	5.9	5.6
SPRS School of Physiotherapy and Rehabilitation Sciences	3	3	100%	84% differs	7	6.2	5.9
All schools	1015	816	80%	The 2024–25 annual report states 81% placed. 816 ÷ 1,015 is 80%.			

CTC in LPA. Eligible and placed counts match the 2024–25 annual report totals (816 placed). A student counts as placed once, however many written offers they hold.

Figure 3. School-wise table. Stated rates that disagree with placed divided by eligible are flagged "differs"; an average above the maximum is flagged "above max".

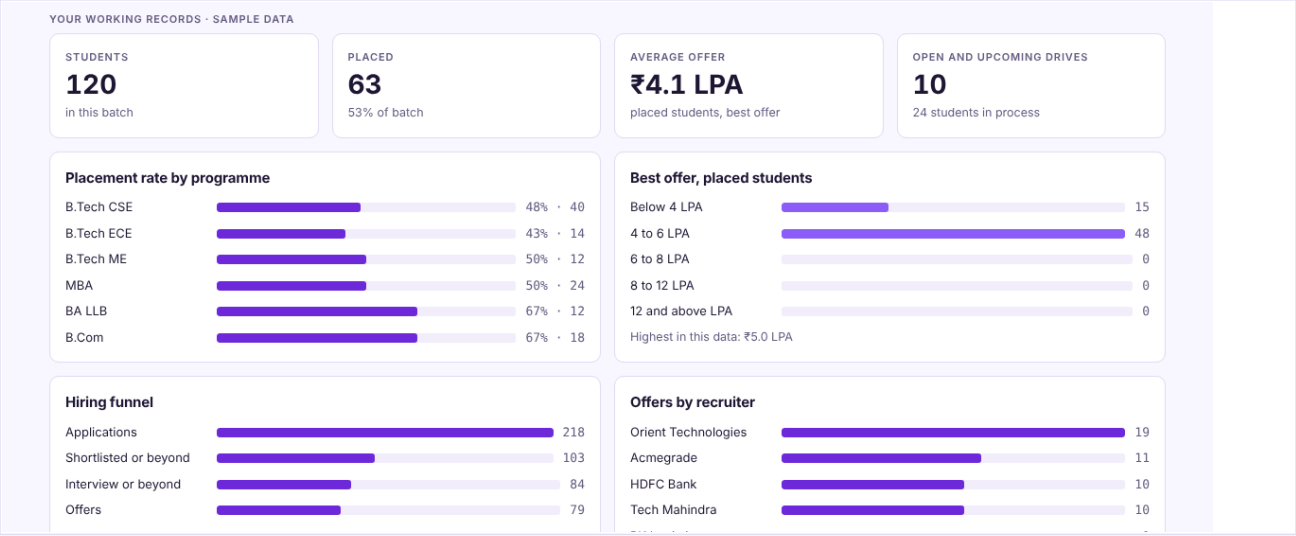


Figure 4. Working-records analytics, computed from the loaded students and applications.

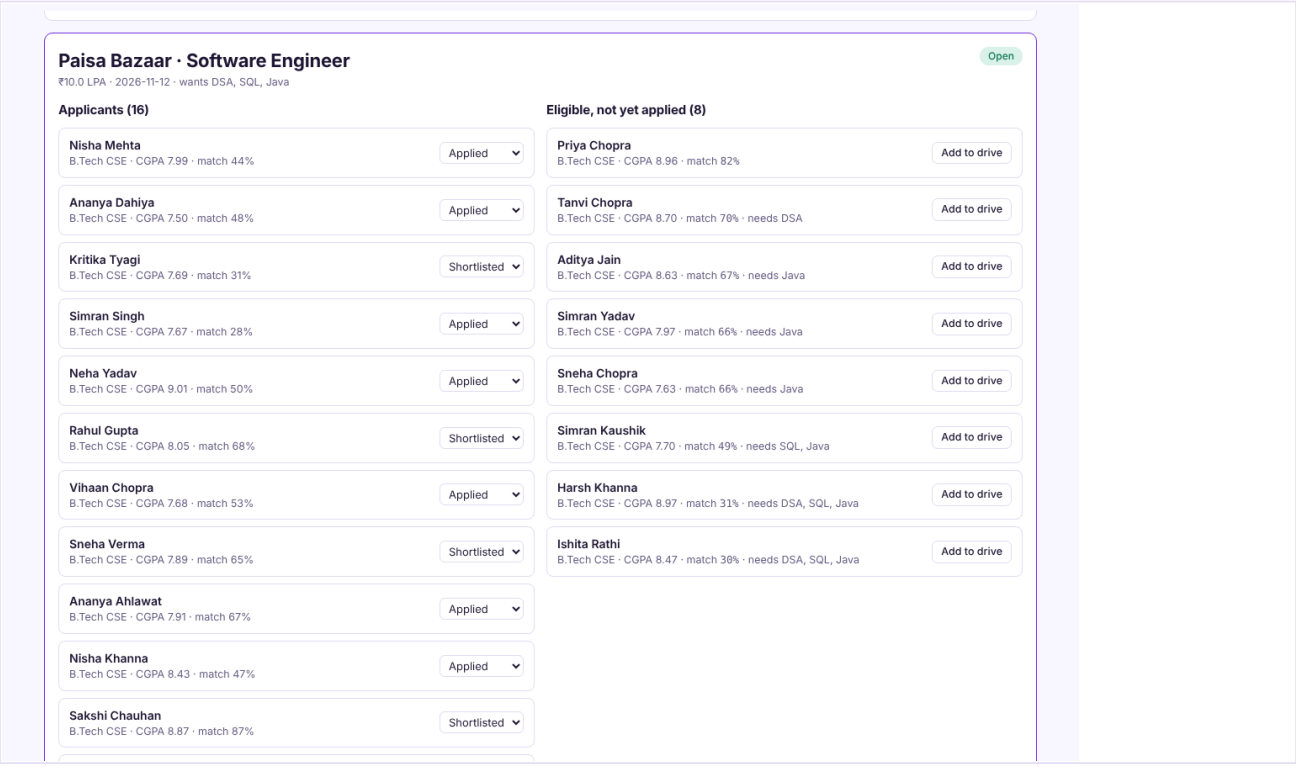
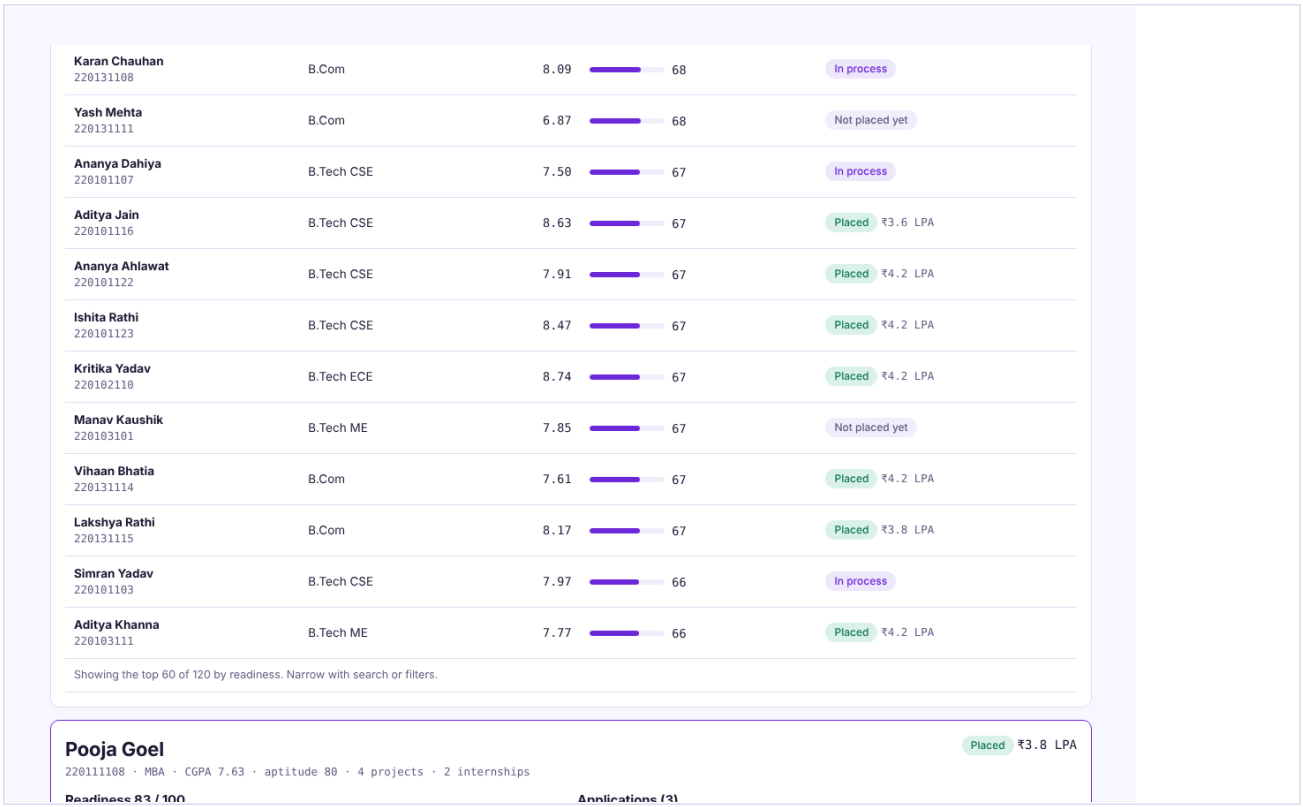


Figure 5. Drives: selecting a drive shows its applicants with stage selectors and the eligible students ranked by match.



Karan Chauhan 220131108	B.Com	8.09	68	In process
Yash Mehta 220131111	B.Com	6.87	68	Not placed yet
Ananya Dahiya 220101107	B.Tech CSE	7.50	67	In process
Aditya Jain 220101116	B.Tech CSE	8.63	67	Placed ₹3.6 LPA
Ananya Ahlawat 220101122	B.Tech CSE	7.91	67	Placed ₹4.2 LPA
Ishita Rathi 220101123	B.Tech CSE	8.47	67	Placed ₹4.2 LPA
Kritika Yadav 220102110	B.Tech ECE	8.74	67	Placed ₹4.2 LPA
Manav Kaushik 220103101	B.Tech ME	7.85	67	Not placed yet
Vihaan Bhatia 220131114	B.Com	7.61	67	Placed ₹4.2 LPA
Lakshya Rathi 220131115	B.Com	8.17	67	Placed ₹3.8 LPA
Simran Yadav 220101103	B.Tech CSE	7.97	66	In process
Aditya Khanna 220103111	B.Tech ME	7.77	66	Placed ₹4.2 LPA

Showing the top 60 of 120 by readiness. Narrow with search or filters.

Pooja Goel — Placed ₹3.8 LPA

220111108 · MBA · CGPA 7.63 · aptitude 80 · 4 projects · 2 internships

Readiness 83 / 100 — Applications (3)

Figure 6. Students: filters, readiness ranking, and a profile with the score breakdown, skills, applications and eligibility reasons.

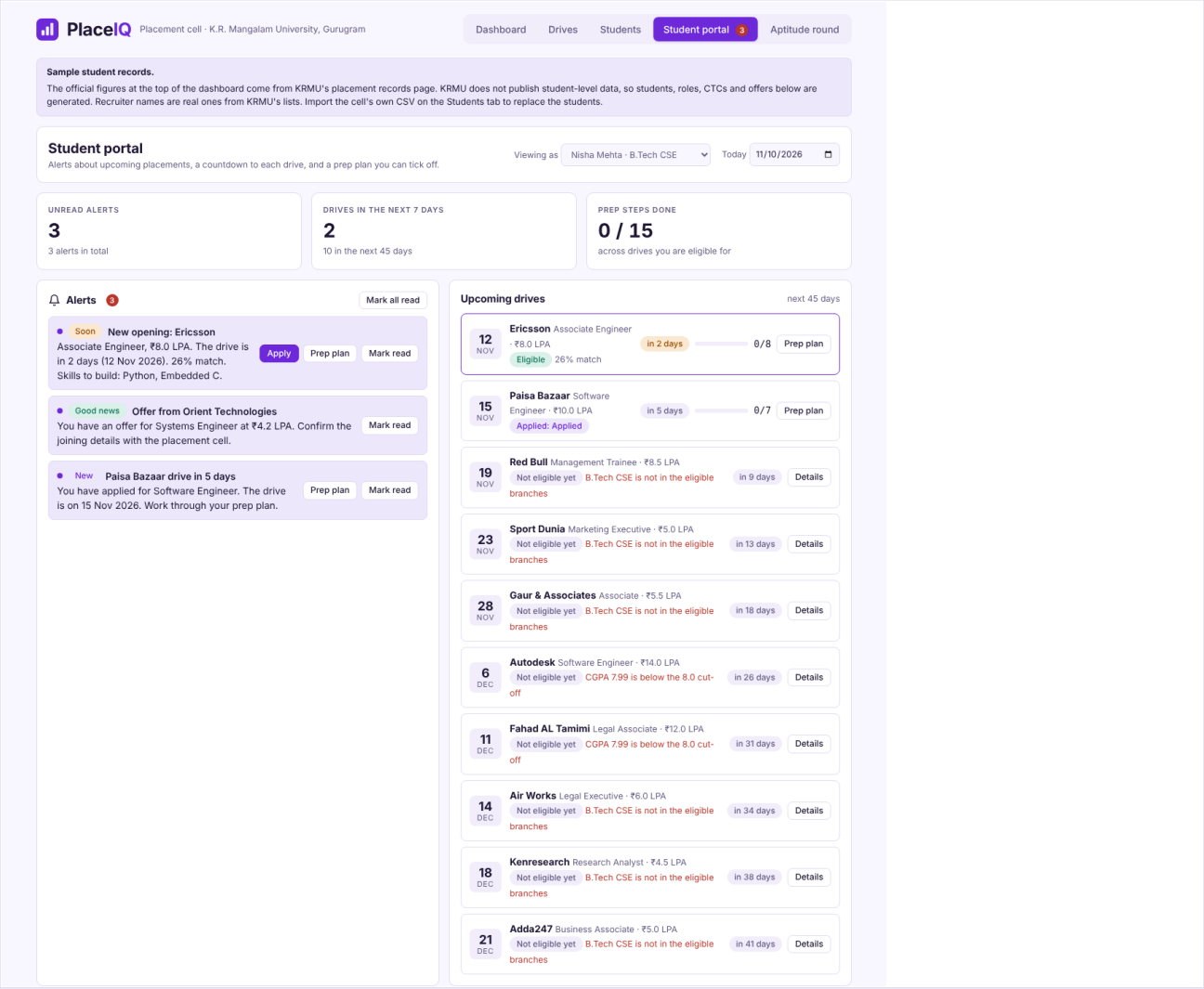


Figure 7. Student portal: unread alerts, upcoming drives with countdowns, and a dated prep plan with tickable steps. The student profile follows below.

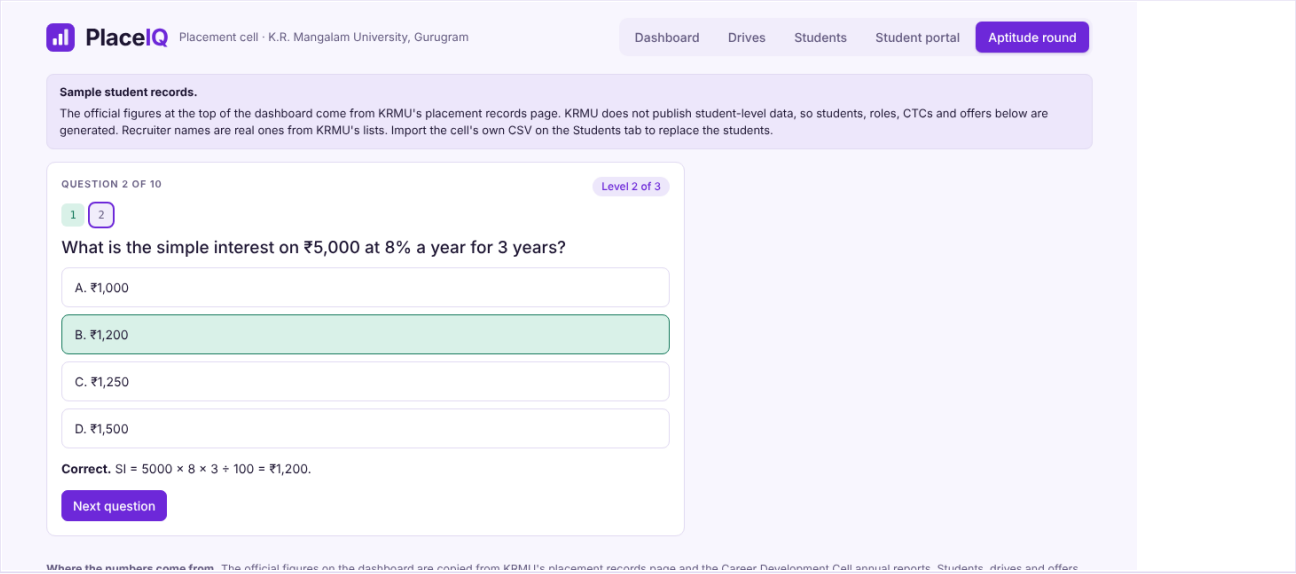


Figure 8. Adaptive aptitude round: level badge, path of answered questions, and instant feedback with an explanation.

Design decisions

- **Official first.** The dashboard opens with the university's own figures, then a divider, then the working records, so a viewer never confuses reported results with sample analytics.
- **Honest data.** Mismatches in the source data are surfaced in the interface. A banner states what is sample on every screen.
- **Identity.** White surfaces on a faint lavender page with one deep violet accent, and a lighter violet for money and package charts. Green, amber and red appear only for status. Bricolage Grotesque for headings, IBM Plex Sans for text, IBM Plex Mono for numbers so columns line up. All colours are tokens in css/tokens.css, so the theme can be changed in one file.
- **Light and dark.** Every colour is a token with a light and a dark value (white and violet, and deep purple); the app follows the device setting. Main text, accent and status colour pairs were checked against WCAG AA contrast in both themes.
- **Phone-ready.** Layouts stack to one column, tables scroll inside their own container, and the page never scrolls sideways at 390 px width.
- **Accessible basics.** Visible keyboard focus, table rows reachable and operable by keyboard, labelled controls, and a polite live region for confirmation messages.
- **State encoded in form.** Placed, In process, Open, Upcoming and Completed use coloured pills, and stage changes confirm with a toast.

7. Testing

The repository has two test suites. 49 unit tests (Node built-in runner, nothing to install) cover the eligibility rules, readiness and match scores, the adaptive quiz engine, CSV import and export, the sample generator's invariants, and the official data: the school totals add up to 1,015 eligible and 816 placed, and the claim that seven of eleven schools have mismatched rates is checked against the data. A browser smoke test (Playwright driving Chrome, 25 checks, all passing) performs these actions and reads back results:

Step	Check	Result
Load dashboard	No page errors; no horizontal overflow at 1280 px	Passed
Drives	Open Paisa Bazaar, add an eligible student to the drive	Passed; student moved from "eligible" to "applicants"
Students	Search, filter by programme, open a profile	Passed

Step	Check	Result
Student portal	Alerts and upcoming drives listed; unread badge on the tab; open a prep plan, tick a step (progress changes); mark all read (badge clears); change the demo date and reset it	Passed
Student portal	Apply to a matched drive, change stage to Offered	Passed; status changed to Placed
Aptitude round	Answer 10 adaptive questions to the result screen	Passed; score written to the profile
CSV import	Import one valid and one invalid row	Passed; "Imported 1 students, skipped 1 invalid row"
Phone layout	All five tabs at 390 px width	No horizontal overflow on any tab
Dark theme	Dashboard rendered with a dark colour scheme	Readable; accents and tokens switch correctly

Findings fixed during testing: the sample-data banner squeezed its text on phones, chart values such as "83% - 266/321" wrapped onto two lines, funnel labels were truncated, and one data-quality note understated how many schools had mismatched rates. All were corrected and the run repeated.

Limits of testing: the only console message was a blocked request for web fonts, because the test environment cannot reach the font host; the layout falls back to system fonts in that case. No tests were run in Safari or Firefox. The GitHub Actions workflow in the repository has not yet run on GitHub itself.

8. Deployment on Vercel or any static host

The reference app is hosted on Vercel. PlaceIQ is a static site, so it deploys the same way. The app uses native ES modules, which browsers do not load from a file:// address, so locally it must be served (npm start, or python3 -m http.server). It needs internet only for the optional Google Fonts.

Option A: drag and drop

- Push the repository to GitHub, then at vercel.com/new import it. Set the framework preset to Other, leave the build command empty and the output directory as the root.
- Every later push redeploys automatically. Vercel gives a public address.

Option B: command line

```
cd placeiq
npx vercel --prod
```

Option C: GitHub

Push the folder to a repository, import it in Vercel, and leave the framework preset as "Other" with no build command and the root as the output directory. Every push then redeploys. GitHub Pages and Netlify work the same way.

Customising the data

- **Official figures:** edit js/data/official.js.
- **Drives:** edit js/data/drives.js, or add drives from the Drives tab.
- **Students:** paste a CSV on the Students tab. Columns: name, roll, branch, cgpa, backlogs, skills (separated by semicolons), aptitude, projects, internships. The last three are optional.
- **Questions:** add objects in js/data/questions.js with a level (1 to 3), options, the index of the correct option and an explanation.
- **Readiness and match weights:** edit READINESS_WEIGHTS and MATCH_WEIGHTS in js/config.js.

9. Limitations and the path to production

Current limitations

- **No shared storage.** Data is saved in one browser on one device. Two officers, or a student on their own phone, would not see each other's changes.
- **No sign-in or roles.** The student portal is a view selector for demonstration, not a secure login. Anyone with the file can see every student.
- **Sample student records.** Until the CDC imports real data, student-level figures are illustrative and must not be quoted as KRMU results.
- **Official figures are typed in.** They will not update when the university updates its page. They reflect the page as read on 31 October 2026.
- **Scoring is rule-based.** The weights are reasonable defaults, not validated against outcomes.
- **Small question bank.** 18 questions is enough to show adaptation; a real round needs hundreds, tagged by topic.

Recommended production architecture

Concern	Recommendation
Front end	Keep the current design; move to React or Next.js if the team wants components and routing, reusing the CSS tokens
API and database	Node.js with a REST or GraphQL API over PostgreSQL; tables for students, drives, applications, offers, assessments and audit log
Authentication	University single sign-on (for example Google Workspace or Microsoft Entra) with roles: student, coordinator, CDC head, recruiter
Data import	Excel and CSV upload with validation and a preview step; scheduled sync from the student information system
Notifications	Email and WhatsApp alerts for new drives, shortlists and interview slots
Recruiter access	Recruiter login to post drives, view eligible shortlists and record outcomes
Assessments	Large tagged question bank, item-response-theory scoring, timed sections, proctoring where required
Analytics	Offers by company, sector and school over several years, with the same discrepancy checks applied to incoming data
Privacy and security	Student records are personal data: apply the Digital Personal Data Protection Act 2023, store consent, limit access by role, encrypt in transit and at rest, keep an audit trail
Operations	Automated tests for scoring and eligibility, staging environment, nightly backups

Suggested next steps

- Ask the CDC for a consented, anonymised export (programme, CGPA band, skills, offers) to replace the sample students and check the readiness weights against real outcomes.
- Reconcile the data problems listed in section 3 with the CDC, then update the figures.
- Decide whether a student-facing login is needed; if so, plan the backend before adding it.

Appendix A. Sources

Source	Used for
KRMU placement records: krmangalam.edu.in/placements/records	Yearly medians, school-wise table, sector split, recruiter lists, headline figures, links to reports
KRMU CDC annual report 2024-25: krmangalam.edu.in/pdfs/placements/cdc-annual-report-2024-25.pdf	Placed counts by school, 1,015 eligible, 816 placed, 81%, recruiter names
KRMU CDC annual report 2025-26: krmangalam.edu.in/pdfs/placements/cdc-annual-report-2025-26.pdf	84% placement, 1200+ offers, 500+ PPOs, 3000+ internship offers, package distribution, 800+ recruiters
AdaptIQ: adaptive-quiz-tool.vercel.app	Reference application: title, description, theme colour (page metadata only)

Appendix B. Files delivered

Path	Purpose
index.html	App shell: header, tabs, mount points, script entry
css/	tokens.css (colours, fonts, light and dark), base.css, layout.css, components.css
js/main.js, app.js, events.js	Entry point; rendering and hash routing; all user interaction
js/config.js	Constants: storage key, stages, programmes, score weights
js/data/	official.js (KRMU figures), drives.js, questions.js, names.js
js/core/	scoring.js, quiz.js, csv.js, sample.js, selectors.js, store.js, util.js: logic with no DOM access
js/ui/, js/views/	Shared HTML builders; one module per tab (dashboard, drives, students, portal, practice)
tests/	unit/ (49 tests) and e2e/smoke_test.py (25 browser checks)
docs/	ARCHITECTURE.md, DATA.md, this report, screenshots
.github/workflows/ci.yml	Runs both test suites on every push
package.json, vercel.json, LICENSE, CONTRIBUTING.md	Scripts only (no dependencies), static hosting config, MIT licence, contributor guide

Appendix C. Glossary

Term	Meaning
CDC	Career Development Cell, KRMU's placement office
CTC / LPA	Cost to company, in lakh rupees per annum (1 lakh = 100,000)
PPO	Pre-placement offer, an offer made after an internship
School codes	SOET Engineering and Technology; SOMC Management and Commerce; SOLS Legal Studies; SOLA Liberal Arts; SMAS Medical and Allied Sciences; SOED Education; SBAS Basic and Applied Sciences; SOAD Architecture and Design; SJMC Journalism and Mass Communication; SOAS Agricultural Sciences; SPRS Physiotherapy and Rehabilitation Sciences
Adaptive test	A test that chooses each next question from the answer to the previous one